

CONFIDENTIAL – INTERNAL USE

SECURITY INCIDENT INVESTIGATION REPORT

Suspected Trojan Activity on a Windows Endpoint

Organization

Cybug Solution

Incident Date

11 September 2026

Prepared By

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Report Date

18 September 2026

Role

SOC / Incident Response

— CYBUG SOLUTION

| DOCUMENT CONTROL

1. Document status

Document Version	1.0
Document Status	Final
Review Status	Pending Review

2. REVISION HISTORY

Version	Date	Change Description
1.0	18 September 2026	Initial incident investigation report

3. APPROVAL STATUS

Approval Stage	Status
Technical Review	Pending
Management Review	Pending
Final Approval	Pending

4. DISTRIBUTION

This report is intended for authorized personnel involved in cybersecurity, incident response, technical investigation, management, and relevant incident stakeholders

EXECUTIVE SUMMARY

1. INCIDENT OVERVIEW

The organization identified a security concern involving a suspicious ZIP file received via WhatsApp from an unknown third-party source. The archive appeared to contain a video file which was subsequently interacted with by an employee on three separate computers. While two endpoints failed to play the file, the primary affected Windows endpoint successfully interacted with the contents. On 11 September 2026, it was observed that the suspicious ZIP file was automatically transmitted through the company's active WhatsApp session to other contacts without the operator's intentional involvement.

2. INITIAL ASSESSMENT

A separate analysis conducted by the VAPT/Bug-Bounty team classified the suspicious artifact as a Trojan. It is important to note that this classification is based entirely on the independent VAPT findings; the SOC investigation did not establish a specific malware family name or reclassify the artifact within the scope of the log analysis.

3. SOC INVESTIGATION

The SOC performed a review of the available Windows Security Event ID 4688 process-creation data using Splunk Enterprise. The analysis focused on identifying potentially suspicious process execution within the available telemetry. Key findings include: analysis of approximately 429 Event ID 4688 records; observations were limited to normal Windows system processes; no matches were identified during searches for commonly abused Living off the Land Binaries (LOLBins); no process execution was detected from the common user-writable paths searched during the investigation; and the imported dataset contained no populated Process_Command_Line values, limiting visibility into specific process parameters

4. Investigation Limitations

At the time of the incident, Sysmon and the Splunk Universal Forwarder were not installed on the affected endpoint. Consequently, the investigation relied primarily on Event ID 4688 logs. Due to these telemetry gaps, the SOC could not conclusively determine if the suspicious artifact successfully executed or establish a direct causal link between a specific process and the unauthorized WhatsApp forwarding behavior.

INCIDENT OVERVIEW

SECTION 1 — INCIDENT SCOPE

Incident Type	Suspected malware / Trojan-related activity
Initial Vector	Suspicious ZIP received through WhatsApp
Affected Environment	Windows endpoint
Artifact	ZIP archive containing an apparent video file
Computers Involved	3
Observed Behavior	ZIP automatically forwarded to other WhatsApp contacts
VAPT Assessment	Artifact separately classified as Trojan

SECTION 2 — INVESTIGATION SCOPE

The SOC investigation focused on available Windows Security Event ID 4688 process-creation logs from the affected endpoint. The evidence was analyzed using Splunk Enterprise to review process activity and identify potentially suspicious execution.

SECTION 3 — ENDPOINT SECURITY CONTROL STATUS

During verification, Windows Security showed Real-time Protection and Cloud-delivered Protection enabled, while Automatic Sample Submission was disabled. Windows Firewall indicated that no action was required. The endpoint had previously had Microsoft Defender protections disabled due to compatibility requirements for offline software. The exact protection state at the time the suspicious file was initially opened could not be conclusively established from the available evidence.

INCIDENT TIMELINE

Purpose: Present the known incident events in chronological order. This timeline records factual events and sequence supported by available data.

DATE		DESCRIPTION
09 Sep 2026	Suspicious File Received	Employee received a suspicious ZIP file through WhatsApp from an unknown/third-party source.
09 Sep 2026	File Opened	The ZIP appeared to contain a video file. The employee opened the ZIP and attempted to interact with the video.
09 Sep 2026	Multiple Computers Involved	The suspicious artifact was opened on three computers. Two computers could not properly play the file, while one primary Windows endpoint was able to interact with/open it.
09 Sep 2026	WhatsApp Session	The company's WhatsApp account was logged into the primary affected Windows endpoint.
11 Sep 2026	Suspicious Forwarding Observed	The same ZIP file was automatically sent through the company's WhatsApp session to other contacts without intentional user action.
Post-Incident	Initial Safety Measures	The client was advised not to reopen or execute the suspicious file and to keep Windows Firewall and Virus & Threat Protection enabled.
Investigation Phase	SOC Log Analysis	Available Windows Security Event ID 4688 evidence was exported and analyzed using Splunk Enterprise.

EVIDENCE COLLECTED

SECTION 1 — EVIDENCE INVENTORY

ID	Source	Description	Purpose
E-01	Windows Security EVTX	Exported Windows Security Event Log containing Event ID 4688 process-creation events	Process creation analysis
E-02	Splunk Analysis Dataset	EVTX-derived Event ID 4688 data imported into Splunk Enterprise	Search, filtering and process activity analysis
E-03	VAPT Assessment	Separate VAPT/Bug-Bounty assessment of the suspicious artifact	Reference only

SECTION 2 — AVAILABLE TELEMETRY

The primary endpoint evidence available for the SOC investigation consisted of Windows Security Event ID 4688 process-creation events. The exported evidence was analyzed in Splunk Enterprise.

INVESTIGATION METHODOLOGY

SECTION 1 — APPROACH

The investigation used a structured SOC analysis approach. Available Windows Security Event Log evidence was extracted, structured, analyzed in Splunk Enterprise, and reviewed for potentially suspicious process activity.

SECTION 2 — INVESTIGATION PROCESS

1. **Evidence Collection** — Review available Windows Security Event Log evidence.
2. **Evidence Extraction** — Extract Event ID 4688 process-creation records into JSON.
3. **SIEM Analysis** — Import the dataset into Splunk Enterprise.
4. **Process Analysis** — Review process names, parent processes, and command-line fields.
5. **Threat Hunting** — Search for LOLBins and execution from selected user-writable paths.
6. **Date-Based Review** — Analyze process activity around 09 and 11 September 2026.
7. **Evidence Assessment** — Determine what the available telemetry supports.

SECTION 3 — ANALYSIS FOCUS

Analysis Area	Objective
Process Creation	Review newly created processes
Parent-Child Relationships	Review process parent-child relationships
Command Line	Check available command-line data
LOLBin Activity	Search for commonly abused Windows utilities
User-Writable Paths	Check selected Downloads, AppData, Temp and Desktop paths
Incident Dates	Review 09 and 11 September 2026 activity

SOC / SPLUNK INVESTIGATION

Dataset Overview

PURPOSE

Document the available Event ID 4688 dataset and establish the starting point for the technical investigation.

SECTION 1 — DATASET OVERVIEW

The exported Windows Security Event Log evidence was analyzed in Splunk Enterprise. The available dataset contained 429 Event ID 4688 process-creation events from the investigated Windows endpoint.

SECTION 2 — DATASET DETAILS

Field	Value
Data Source	Windows Security Event Log
Event ID	4688
Event Type	Process Creation
Total Events	429
Splunk Source	4688_eventfile (1).evtx
Sourcetype	WinEventLog:Security
Host	DESKTOP-98BU7RN

SECTION 3 — INVESTIGATION NOTE

"These event counts represent the available Event ID 4688 dataset and do not by themselves establish malicious or benign activity."

SECTION 4 — INITIAL PROCESS INVENTORY

The initial process inventory showed a limited set of process names, which were subsequently reviewed individually and by incident date.

Process	Events
smss.exe	117
csrss.exe	78
autochk.exe	39
lsass.exe	39
services.exe	39
wininit.exe	39
winlogon.exe	39
Registry	39

New Search Save As Create Table View Close

source="4688_eventfile (1).evtx" EventCode=4688
 | stats count by New_Process_Name
 | sort - count

Time range: All time Q +

✓ 429 events (before 9/16/26 3:26:29.000 PM) No Event Sampling Job || → 🖨 ⬇ 💡 Smart Mode

Events Patterns **Statistics (8)** Visualization

Show: 20 Per Page Format Preview: On

New_Process_Name	count
C:\Windows\System32\smss.exe	117
C:\Windows\System32\csrss.exe	78
C:\Windows\System32\autochk.exe	39
C:\Windows\System32\lsass.exe	39
C:\Windows\System32\services.exe	39
C:\Windows\System32\wininit.exe	39
C:\Windows\System32\winlogon.exe	39
Registry	39

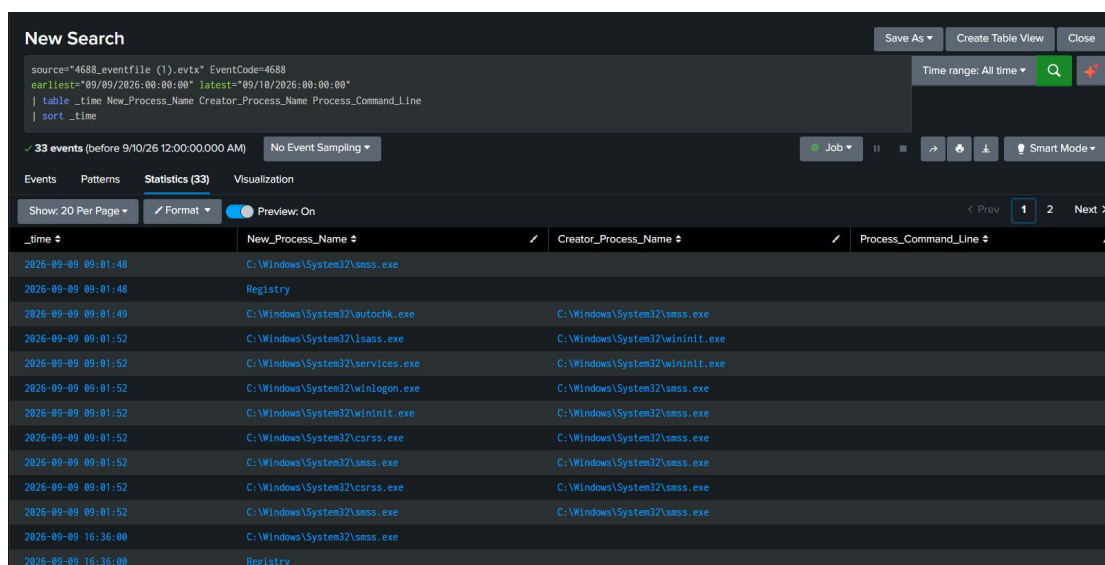
Figure 2 — Splunk analysis of the available Event ID 4688 process-creation dataset.

09 SEPTEMBER 2026 — PROCESS ACTIVITY ANALYSIS

SECTION 1 — SEARCH SCOPE

Windows Security Event ID 4688 process-creation events for 09 September 2026 were reviewed in Splunk Enterprise to examine recorded process activity during the incident date.

SECTION 2 — EVIDENCE



New Search

source="4688_eventfile (1).evtx" EventCode=4688
earliest="09/09/2026:00:00:00" latest="09/10/2026:00:00:00"
| table _time New_Process_Name Creator_Process_Name Process_Command_Line
| sort _time

Time range: All time

33 events (before 9/10/26 12:00:00.000 AM) No Event Sampling

Events Patterns Statistics (33) Visualization

Show: 20 Per Page Format Preview: On

_time	New_Process_Name	Creator_Process_Name	Process_Command_Line
2026-09-09 09:01:48	C:\Windows\System32\smss.exe		
2026-09-09 09:01:48	Registry		
2026-09-09 09:01:49	C:\Windows\System32\autochk.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\lsass.exe	C:\Windows\System32\wininit.exe	
2026-09-09 09:01:52	C:\Windows\System32\services.exe	C:\Windows\System32\wininit.exe	
2026-09-09 09:01:52	C:\Windows\System32\winlogon.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\wininit.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\csrss.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\smss.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\csrss.exe	C:\Windows\System32\smss.exe	
2026-09-09 09:01:52	C:\Windows\System32\smss.exe	C:\Windows\System32\smss.exe	
2026-09-09 16:36:00	C:\Windows\System32\smss.exe		
2026-09-09 16:36:00	Registry		

Figure 3 — Splunk review of Event ID 4688 process activity recorded on 09 September 2026.

SECTION 3 — ANALYSIS

The available events showed standard Windows system processes and expected parent-child relationships. No suspicious process name was identified in the reviewed 09 September Event ID 4688 records.

11 SEPTEMBER 2026 — PROCESS ACTIVITY ANALYSIS

SECTION 1 — SEARCH SCOPE

Windows Security Event ID 4688 process-creation events for 11 September 2026 were reviewed in Splunk Enterprise to examine the process activity recorded on the date when automatic forwarding of the suspicious ZIP was observed.

SECTION 2 — EVIDENCE

The screenshot shows the Splunk Search interface with a search query for Event ID 4688. The search results are displayed in a table with columns: _time, New_Process_Name, Creator_Process_Name, and Process_Command_Line. The results show a sequence of process creations starting at 09:10:37, including Registry, autochk.exe, csrss.exe, lsass.exe, services.exe, winlogon.exe, wininit.exe, and csrss.exe, all created by smss.exe.

_time	New_Process_Name	Creator_Process_Name	Process_Command_Line
2026-09-11 09:10:37	C:\Windows\System32\smss.exe		
2026-09-11 09:10:37	Registry		
2026-09-11 09:10:38	C:\Windows\System32\autochk.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:43	C:\Windows\System32\csrss.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:43	C:\Windows\System32\lsass.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:44	C:\Windows\System32\lsass.exe	C:\Windows\System32\wininit.exe	
2026-09-11 09:10:44	C:\Windows\System32\services.exe	C:\Windows\System32\wininit.exe	
2026-09-11 09:10:44	C:\Windows\System32\winlogon.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:44	C:\Windows\System32\wininit.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:44	C:\Windows\System32\csrss.exe	C:\Windows\System32\smss.exe	
2026-09-11 09:10:44	C:\Windows\System32\smss.exe	C:\Windows\System32\smss.exe	

Figure 4 — Splunk review of Event ID 4688 process activity recorded on 11 September 2026.

SECTION 3 — ANALYSIS

The available events consisted of standard Windows system processes with expected parent-child relationships. No suspicious process name was identified in the reviewed 11 September Event ID 4688 records.

SUSPICIOUS ACTIVITY HUNTING

SECTION 1 — PURPOSE

Targeted searches were performed against the available Event ID 4688 process-creation dataset to identify commonly abused Windows utilities, execution from selected user-writable locations, and available process command-line activity.

SECTION 2 — LOLBIN / SUSPICIOUS UTILITY SEARCH

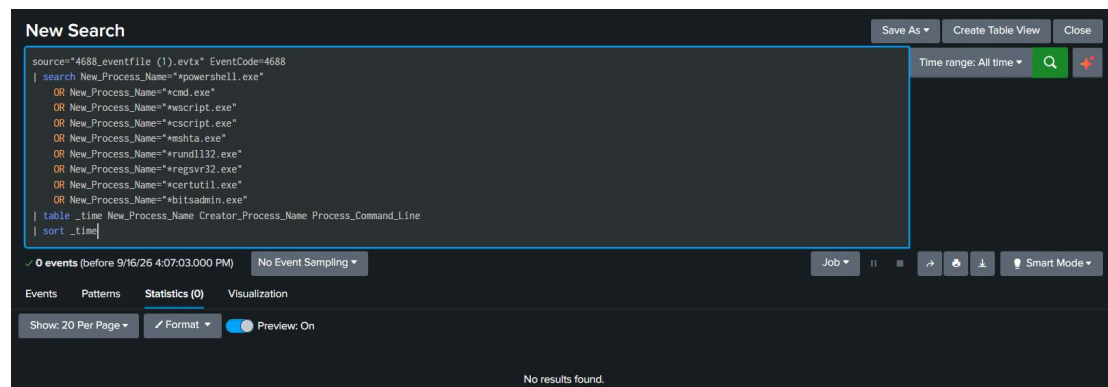


Figure 5 — Splunk search for commonly abused Windows utilities in the available Event ID 4688 dataset.

Result: No matching events were returned.

SECTION 3 — USER-WRITABLE PATH SEARCH

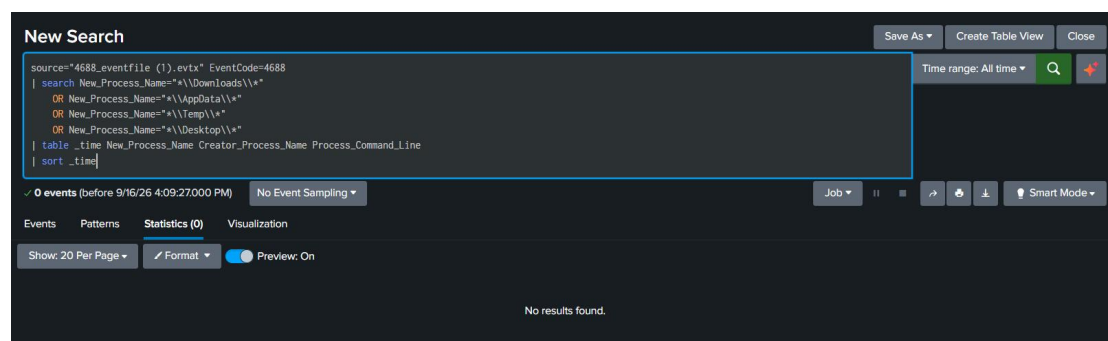


Figure 6 — Splunk search for process execution from selected user-writable paths.

Result: No matching process-creation events were returned.

SECTION 4 — COMMAND-LINE AVAILABILITY

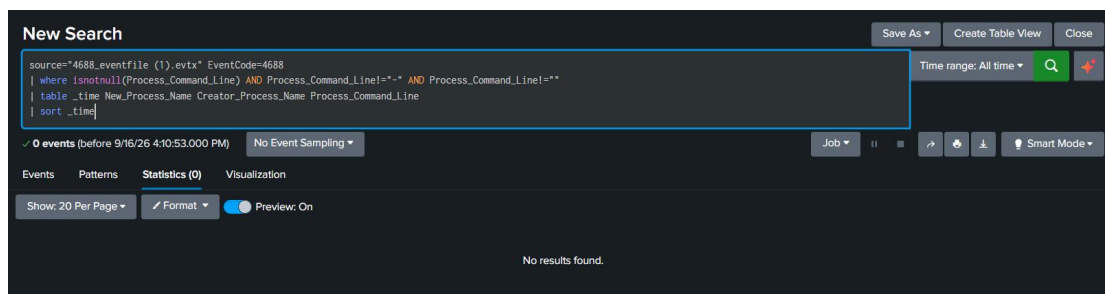


Figure 7 — Splunk assessment of available process command-line information.

Result: No populated Process_Command_Line values were present in the imported dataset.

SECTION 5 — ANALYSIS

The targeted searches did not identify matching events for the searched LOLBins or selected user-writable execution paths, and no populated process command-line values were available in the imported dataset. These results are limited to the telemetry available in the Event ID 4688 evidence.

INVESTIGATION FINDINGS & ASSESSMENT

SECTION 1 — PROCESS ACTIVITY

Analysis of the available Windows Security Event ID 4688 data showed standard Windows system processes and expected parent-child relationships on the reviewed incident dates. No suspicious process name was identified in the available process-creation records.

SECTION 2 — SUSPICIOUS ACTIVITY HUNTING

Targeted searches for commonly abused Windows utilities (LOLBins), execution from selected user-writable paths, and populated process command-line values returned no matching results in the available dataset.

SECTION 3 — KEY FINDINGS

Finding	Assessment
Event ID 4688 Dataset	429 process-creation events were available for analysis.
09 September Activity	Standard Windows system processes were observed.
11 September Activity	Standard Windows system processes were observed.
LOLBin Search	No matching events identified.
User-Writable Path Search	No matching events identified.
Command-Line Data	No populated Process_Command_Line values available.

SECTION 4 — OVERALL ASSESSMENT

Based on the available Event ID 4688 evidence, the SOC investigation did not identify suspicious process-creation activity associated with the reviewed dates or the targeted searches. However, the available telemetry was limited and does not by itself establish that the endpoint was free of malicious activity or conclusively determine whether the suspicious artifact executed.

MALWARE ASSESSMENT REFERENCE

SECTION 1 — MALWARE ASSESSMENT

The suspicious ZIP/artifact was separately analyzed by the VAPT/Bug-Bounty team and classified as a Trojan. The detailed malware assessment was performed separately and is referenced here as supporting incident information.

SECTION 2 — RELATION TO SOC INVESTIGATION

The SOC investigation used this assessment as incident context while independently analyzing the available Windows endpoint telemetry. The SOC analysis focused on Event ID 4688 process-creation activity and targeted searches within the available dataset.

SECTION 3 — RESPONSIBILITY

Area	Activity
Malware Assessment	Analysis and classification of the suspicious artifact
SOC / Incident Response	Investigation of available endpoint telemetry

SECTION 4 — REFERENCE NOTE

"The malware assessment and SOC investigation are complementary. The malware classification is attributed to the separate assessment, while the findings presented in this report are based on the endpoint evidence available to the SOC."

RECOMMENDATIONS

PURPOSE: Present practical security improvements and management suggestions based strictly on the findings, evidence, and limitations identified during the incident investigation.

"Based on the available evidence, investigation findings, and identified limitations in historical endpoint telemetry, the following recommendations are proposed to improve security prevention, visibility, detection, and incident-response capabilities for future security events."

01 ENDPOINT SECURITY PROTECTION

Ensure Windows Firewall and Virus & Threat Protection remain enabled and actively monitored on affected and other relevant endpoints. Security protections should not be disabled for software compatibility without an approved security exception and an appropriate alternative control.

02 WINDOWS SECURITY CONFIGURATION REVIEW

Review and maintain Windows Security settings, including Real-time Protection, Cloud-delivered Protection, Automatic Sample Submission, and other applicable endpoint security controls, in accordance with the organization's approved security policy.

03 ENDPOINT MONITORING & TELEMETRY

Consider implementing appropriate endpoint monitoring and security-log collection to provide greater visibility into process activity and other relevant security events.

04 CENTRALIZED SECURITY MONITORING

Consider centralizing relevant endpoint and security logs within an approved SIEM/SOC monitoring platform to support correlation, investigation, and alerting.

05 DETECTION & ALERTING

Develop and maintain detection rules for suspicious authentication activity, abnormal process execution, potentially abused Windows utilities, and other security behaviors relevant to the organization's environment.

06 SECURITY AWARENESS

Strengthen employee awareness regarding suspicious ZIP files, unexpected attachments, unknown senders, and potentially malicious files received through messaging platforms. Employees should be encouraged to report suspicious files rather than opening or forwarding them.

07 INCIDENT RESPONSE PROCEDURE

Establish or strengthen a documented procedure for reporting, triaging, containing, investigating, and escalating suspected malware or endpoint security incidents.

08 PRODUCTION CHANGE CONTROL

Any deployment of security agents, monitoring tools, configuration changes, or other controls on production infrastructure should undergo appropriate technical evaluation, authorization, and change-management procedures before implementation.

Recommendations are based on the available evidence and investigation scope and should be evaluated against the client's approved security architecture, operational requirements, and change-management procedures.

CONCLUSION

01 INVESTIGATION SUMMARY

The investigation reviewed the available Windows Security Event ID 4688 process-creation telemetry associated with the affected Windows endpoint. The available events were extracted from the supplied EVTX evidence and analyzed using Splunk.

02 KEY OBSERVATION

The reviewed Event ID 4688 data primarily contained standard Windows system processes with expected parent-child relationships. No suspicious process names were identified in the available 09 September and 11 September 2026 process-creation activity.

03 THREAT-HUNTING RESULT

Targeted searches for selected LOLBins, selected user-writable execution paths, and populated process command-line values did not return matching events in the available dataset.

04 LIMITATION s ASSESSMENT

The available telemetry was limited because detailed endpoint monitoring such as Sysmon and Splunk Universal Forwarder was not present at the time of the incident. Therefore, the available evidence does not establish that the endpoint was free from malicious activity and does not conclusively determine whether the suspicious artifact executed successfully.

05 OVERALL CONCLUSION

The suspicious artifact was separately assessed by the VAPT/Bug-Bounty function and classified as a Trojan. The SOC investigation independently assessed the available endpoint telemetry and did not identify corresponding suspicious process-creation activity within the reviewed Event ID 4688 dataset. Continued improvement of endpoint visibility, security controls, centralized monitoring, and incident-response processes is recommended for future security events.

APPENDIX / EVIDENCE REFERENCES

SECTION 1 — EVIDENCE REFERENCES

ID	Evidence	Description	Purpose
E-01	4688_eventfile (1).evtx	Windows Security Event Log evidence containing Event ID 4688 process-creation events.	Endpoint process-creation investigation.
E-02	4688.json	Structured JSON extraction of the available Event ID 4688 data.	SIEM-based analysis and threat hunting.
E-03	VAPT / Bug-Bounty Assessment	Separate assessment of the suspicious ZIP/artifact.	Reference for the malware classification provided by the VAPT/ Bug-Bounty function.

SECTION 2 — TECHNICAL ANALYSIS REFERENCES

- Windows Security Event ID 4688 — Process Creation
- Splunk analysis of the available Event ID 4688 dataset
- 09 September 2026 process-activity review
- 11 September 2026 process-activity review
- Selected LOLBin hunting
- Selected user-writable path hunting
- Process command-line availability review

SECTION 3 — INVESTIGATION SCOPE NOTE

"The evidence and analysis presented in this report are limited to the artifacts and telemetry available within the defined investigation scope. Absence of an event within the available dataset should not be interpreted as proof that the corresponding activity did not occur."

SECTION 4 — EVIDENCE HANDLING NOTE

"Evidence referenced in this report was used for investigation and analysis. Original evidence should be retained according to the organization's applicable evidence-handling, access-control, and retention procedures."